BCA (Semester-I) Examination, 2022

(Session : 2022-25)

COMPUTER APPLICATION

[Paper Code : BCA-101]

(Mathematical Foundation)

Time : Three Hours [Maximum Marks : 80]

Note: *Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer **any five** questions.*

1. Find the Eigen value and Eigen vector:

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

2. If $y^{1/m} + y^{-1/m} = 2x(x^2 - 1)$,

prove that $y'' + \frac{1}{x} y' - m^2 y = 0$.

3. Evaluate $I = \int_0^{\pi/2} \sqrt{1 + \sin x}\ dx$

4. If $A = \begin{bmatrix} 2 & 3 \\ -1 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 5 \\ 1 & -2 \end{bmatrix}$, $C = \begin{bmatrix} x & y \\ 8 & 0 \end{bmatrix}$,

find the value of x and y.

If $A + B = C$

5. Solve : $(x^2 - y^2) \frac{dy}{dx} = 2xy$

6. Integrate : $I = \int_0^{\pi/2} \int_0^{x} (x^2 + y^2)\, dy\, dx$

7. Find the inverse of the matrix $A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 3 & 0 \\ 0 & 3 & 4 \end{bmatrix}$

8. Apply Maclaurin's theorem to obtain the expansion of

$\log(1 + t)$.

9. Solve : $\frac{dy}{dx} = \sin 2x$

10. Solve : $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = e^{2x}$

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